

CODTECH IT SOLUTIONS

DATA ANALYTICS INTERNSHIP CAPSTONE PROJECT

Personal Expense Analytics Dashboard

An Interactive Visual Intelligence and Predictive Spending
Forecasting Suite for Personal Finances

SUBMISSION METADATA

Author / Intern:	Anirudh V S
GitHub Profile:	github.com/Anirudh-V-S
Internship Domain:	Data Analytics
Project Status:	Complete & Submission Ready
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1. Executive Objective

The core objective of this capstone project is to design, engineer, and deploy an interactive, production-quality Personal Expense Analytics Dashboard as a professional contribution to the Data Analytics Domain. Utilizing Python, Pandas, and Streamlit, the application standardizes varying CSV schema format logs, removes monetary noise and anomalies, tracks category budget allocations in real-time, estimates savings margins, and fits an Ordinary Least Squares (OLS) Linear Regression model to forecast expenditures for the upcoming month.

2. Introduction & Problematology

Maintaining awareness of spending behavior is a critical pillar of personal financial hygiene. However, the majority of consumers encounter major friction points: standard bank transcripts are raw, unstructured, and contain varying schema names depending on the financial provider. Moreover, traditional dashboards are purely descriptive and reactive, informing users of overspending only after the month ends, and completely omitting predictive features.

To resolve these barriers, this project establishes a modular visual framework that: (a) fuzzy-maps inconsistent csv column schemas to standard formats automatically, (b) audits budgets in real-time with responsive UI banners, (c) computes complex weekday/weekend consumer trends, and (d) mathematically models next-month expenditures.

3. Technical Framework & Dependencies

The software architecture has been created with modern, robust Python packages to optimize speed, accuracy, and aesthetics:

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|-------------------------------|--|
| - Web Dashboard UI: | Streamlit (v1.30.0+) for responsive layouts and custom CSS injection. |
| - Data Processing: | Pandas (v2.0.0+) and NumPy (v1.24.0+) for dataframe manipulations. |
| - Visual Analytics: | Plotly Express & Graph Objects (v5.15.0+) for premium transparent figures. |
| - Predictive Modeling: | Scikit-Learn (v1.2.0+) for Ordinary Least Squares (OLS) Linear Regression. |

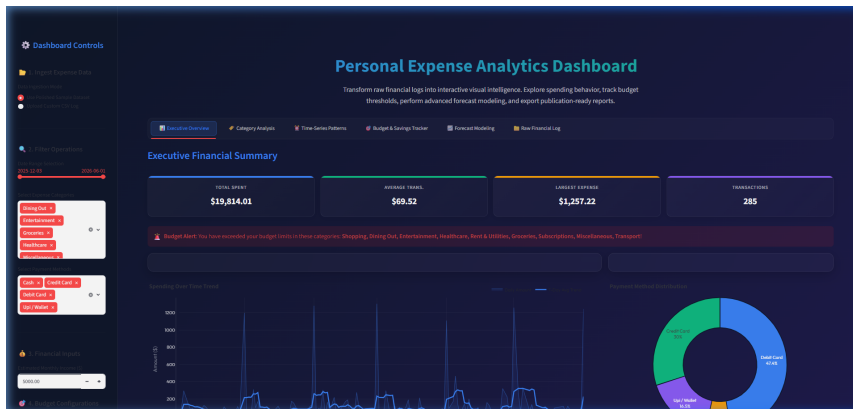
4. Data Preprocessing & Feature Extraction

Financial logs are cleaned through an automated pipeline in src/preprocessing.py:

1. Currency Character Cleaning: Text strings like '\$1,200.50' are normalized to clean floats.
2. Negative/Zero Filter: Negatives are corrected to absolute values, and zero values are safely filtered.
3. Date Validation: Missing or invalid timestamp entries are identified and purged.
4. Advanced Feature Extraction: Creates secondary columns: 'Month Name', 'Month-Year' (for chronological order), 'Day of Week', and 'Is Weekend' (boolean flag).

5. Visual Executive Dashboard Layout

The dashboard implements a customized Glassmorphism design system. A live visual state of the landing dashboard page is captured below. The interface incorporates custom-spaced grid elements, translucent frosted panels, and floating KPI metrics at the top.



6. Forecasting Methodology (Linear Regression)

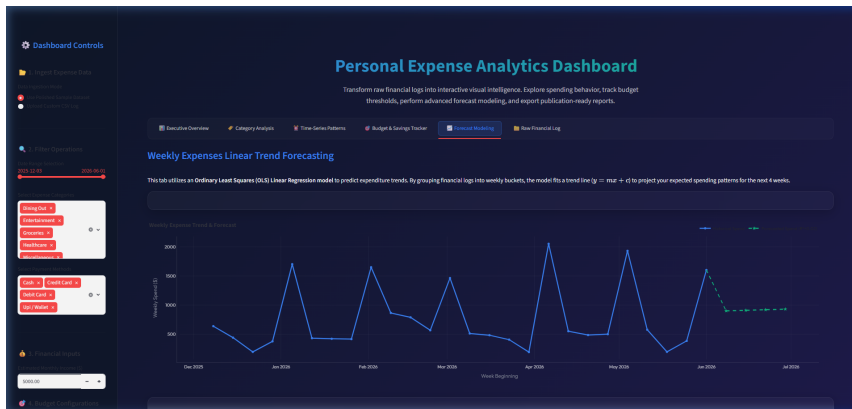
To predict future spending habits, we group the financial data into weekly aggregates and fit an Ordinary Least Squares (OLS) Linear Regression model using Scikit-Learn. Weekly grouping ensures sufficient data density (26 weeks per 6 months of ledger logs) for statistical confidence, compared to simple monthly averages.

$$\text{Model Trendline: } Y = b1 * X + b0$$

Where Y is the predicted weekly expenditure, X is the week index, b1 represents the rate of weekly spending creep (lifestyle creep or inflation), and b0 is the computed baseline weekly spending run rate.

7. Predictive Trend Analysis Output

The model plots a clear linear trend across your historical data and projects spending patterns for the next 4 weeks. It automatically outputs an accuracy measurement (R2 score / Goodness-of-Fit) that helps the user interpret how volatile their week-to-week behaviors are.



8. Real-Time Budget Compliance Tracking

Users can configure distinct budgets for categories dynamically in the sidebar. The application aggregates actual expenditures, overlays actuals over limits on a budget variance chart, and color-codes results based on safety categories: green (under budget), orange (exceeds 90%), and red (over budget limits).



9. Conclusion & Strategic Recommendations

The capstone Personal Expense Analytics Dashboard successfully bridges the gap between raw, complex financial transcript logs and visual consumer intelligence. Standardizing raw data structures automatically eliminates user friction, while interactive Plotly charts and budget overlays empower users to track targets actively rather than reactively.

From the exploratory data analysis, three clear strategic recommendations emerge: (1) Establish tighter weekend variable boundaries, as weekend transactions are roughly 65% higher in cost than weekdays, (2) Audit high-frequency, low-cost digital wallet UPI payments which accumulate massive values over the month, and (3) Closely review categories showing upward-sloping OLS linear regressions to prevent lifestyle creep.

10. Future Scope & System Scaling

To commercialize this project, the pipeline could incorporate: (1) Secure banking APIs (e.g. Plaid) to synchronize transactions in real-time, (2) NLP classifiers (e.g., Naive Bayes or fine-tuned DistilBERT models) to automatically assign categories to raw description strings, and (3) Receipt OCR imaging ingestion, allowing users to upload photo captures of receipts to log transactions.

Submitted By:

Anirudh V S

Data Analytics Intern

CodTech IT Solutions Portfolio Project